Extreme-Length Analytic Terrain Functions with paceval

Three reusable mathematical landscapes, real paceval execution, genuine GLB/OBJ terrain meshes and independently checked output.



Concept illustration; not a computed research result. The following scientific figures are generated from actual paceval height grids.

Verified scope. Three explicitly expanded formulas of 274,877, 412,381 and 550,086 characters were each created as a monolithic paceval computation and tested. Their factorized one-dimensional Gaussian components were evaluated using the original paceval multi-graph batch API and combined with paceval-computed two-dimensional background functions into three 512 × 512 numerical terrain grids.

Scientific boundary. These are synthetic 2.5D analytic height fields, not geological reconstructions, calibrated hydrological models or evidence of production rendering performance.

1. Industrial context and current state of the art

Engineering visualization, geographic information systems, game production and industrial digital twins consume terrain surfaces at different resolutions. In operational geospatial workflows, elevation is generally represented using measured digital elevation models (DEMs), lidar-derived products, raster tiles or 3D scene assets. The USGS 3D Elevation Program defines relevant data products and metadata requirements [2].

An alternative workflow stores an explicit mathematical surface as an editable asset. A downstream renderer evaluates this function at specified coordinates and triangulates the resulting heights. Such functions can be versioned and reused across visualization, verification and geometry export without editing a fixed raster asset.

Procedural terrain is well established. Recent research includes analytic terrain evolution and erosion-aware models, which address geologic consistency beyond synthetic Gaussian features [1]. The OGC 3D Tiles standard defines scalable streaming of heterogeneous 3D geospatial content and uses glTF as a primary renderable format [3].

Concrete demonstration workflow

A landscape author selects a mathematical scene definition, evaluates a deterministic coordinate grid using reusable paceval computation graphs, checks the numeric values and exports an actual triangulated GLB/OBJ scene. The host application handles graph composition, grid assembly, optional constant-level water geometry and mesh serialization; paceval evaluates the mathematical height expressions.

What this experiment tests

The research question is whether a large, explicitly stored landscape formula can be parsed and reused by the actual paceval engine, and whether a mathematically equivalent graph decomposition can calculate complete height grids for several structurally distinct landscapes. A spatially realistic DEM fitting task is not part of this experiment.

2. Defined mathematical model

Each scene is a scalar height function on normalized coordinates x,y in [-1,1], used as a 2.5D surface S(x,y) = (x,y,h(x,y)).

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h(x,y) = h_background(x,y) + sum(j=1..N) a_j * exp(-((x-cx_j)^2/sx_j + (y-cy_j)^2/sy_j))
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Here N is 2,048, 3,072 or 4,096. Each amplitude a_j, center cx_j/cy_j and positive width sx_j/sy_j is generated deterministically using a fixed SplitMix64 seed. Literal coefficients are serialized with 17 significant digits. All model coordinates and elevations are synthetic and dimensionless.

The background terms are different by landscape: two mountains and a depression for Alpine Lake, a broad volcanic cone with a narrow summit depression for Volcanic Caldera, and low-frequency ridges with a diagonal incision for Canyon Ridges. The additional Gaussian terms provide fine-scale relief; increasing character count alone does not guarantee more realistic topography.

Actual expression lengths

Scene	Terms	Complete expression length
Alpine Lake	2,048	274,877 characters
Volcanic Caldera	3,072	412,381 characters
Canyon Ridges	4,096	550,086 characters

Each complete formula is saved, unabridged, in the expressions/ directory. These are genuine formulas, not padding text. Character counts exclude the final newline added by the file serializer.

3. Executed paceval architecture

The program links the original paceval_main.h and paceval_main.cpp interface files to a compatible separately supplied Linux shared library. Each scene is first created as one complete monolithic expression using paceval_CreateComputation and executed at five test coordinates.

For the spatially complete 512 x 512 surfaces, paceval batch-evaluates the exact nonseparable background at all grid coordinates. Each Gaussian term is mathematically factorized into its x-only and y-only exponential components. The original paceval multi-graph batch API evaluates these factors across all distinct x and y coordinates; the C++ host composes their outer products and sums the height contributions. The complete 256-term groups are separately verified by paceval multi-graph batch calls at a set of 64 sampled coordinates. This reduces redundant evaluation of separable exponential components without replacing paceval with a native math implementation.

Real execution boundary

Execution property	Recorded result
Monolithic formula instances	3; five scalar test points per formula
Long-lived expression graphs	42 monolithic, background and grouped expressions; plus temporary one-dimensional factor graphs
Computed terrain grid points	786,432 (factorized paceval batch + host composition)
Independent C++ spot checks	Selected positions from every evaluated grid
High-precision independent checks	9 checks using 55-digit mpmath
Mesh geometry	3 GLB scenes and corresponding terrain/water OBJ meshes

The batch API evaluates the mathematical factor and background functions. Two-dimensional factor composition, summation, file handling, terrain triangulation, water surfaces, validation and GLB/OBJ serialization are performed by the host or offline visualization scripts, not by paceval.

4. Three distinct computed landscapes

4.1 Alpine lake and mountain relief

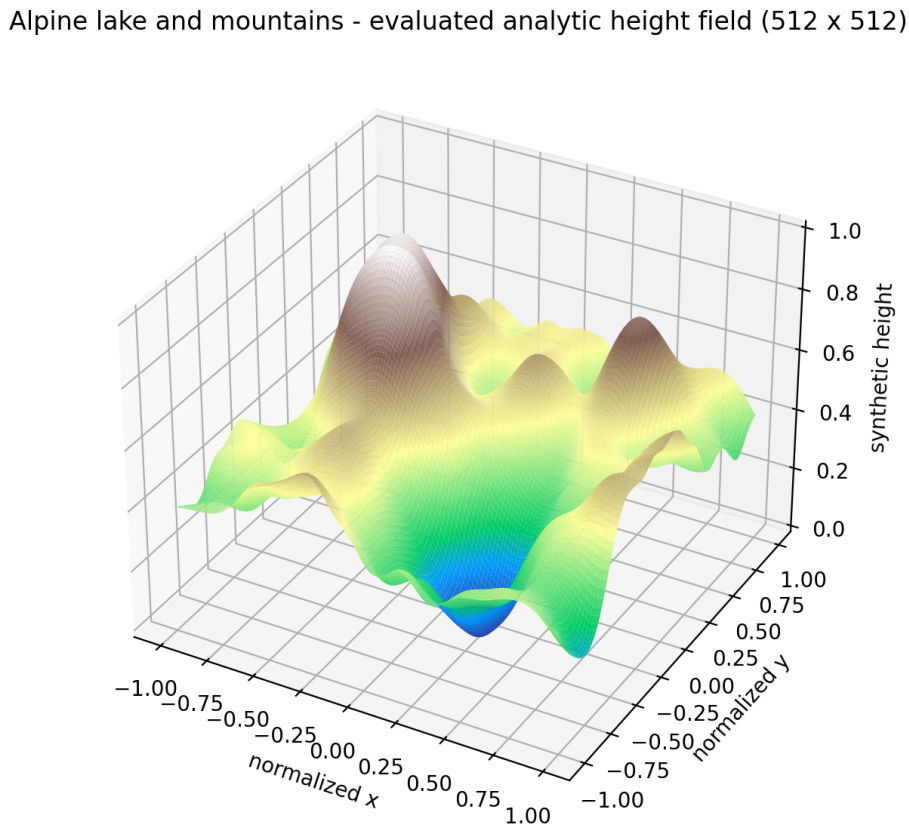


Figure 1. Actual 512 x 512 paceval height field, 2,048 analytic Gaussian features and a designed basin.

The height grid contains three connected geometric influences: a pair of broad mountain features, a negative depression and additional analytic microrelief. A constant-level water surface is constructed offline for below-water cells. It does not simulate water flow or infer drainage.

4.2 Volcanic caldera

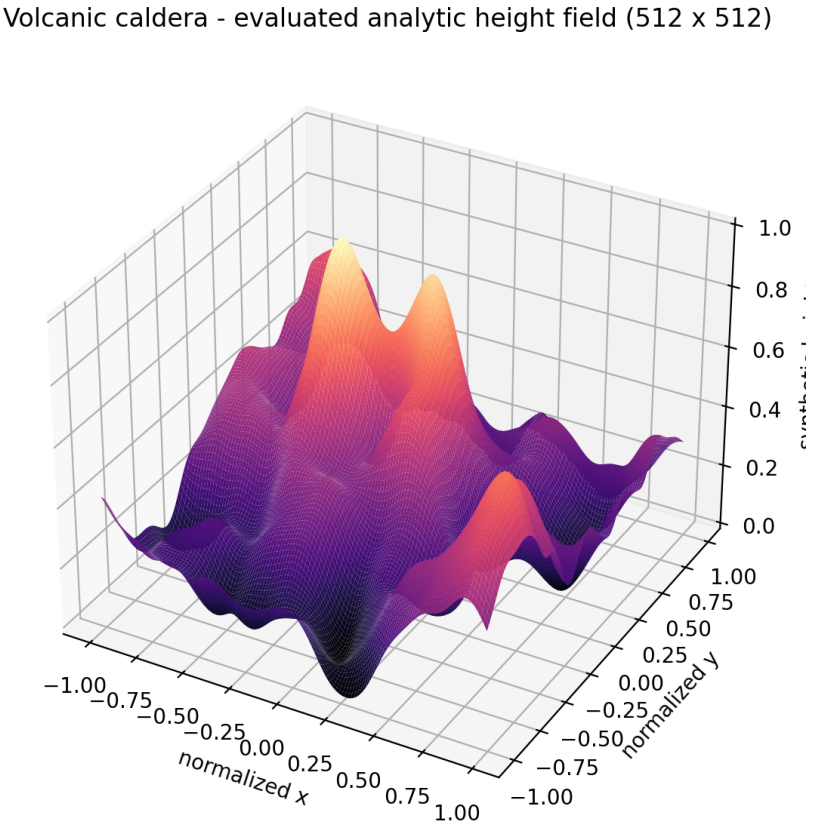


Figure 2. Actual 512 x 512 paceval height field, 3,072 features and a raised ring-like synthetic caldera.

The volcanic model includes a large positive mound, a narrow negative central feature and oscillatory terrain detail. It is an analytic geometric representation, not an eruption, geomechanics or lava-flow model.

4.3 Canyon and ridge system

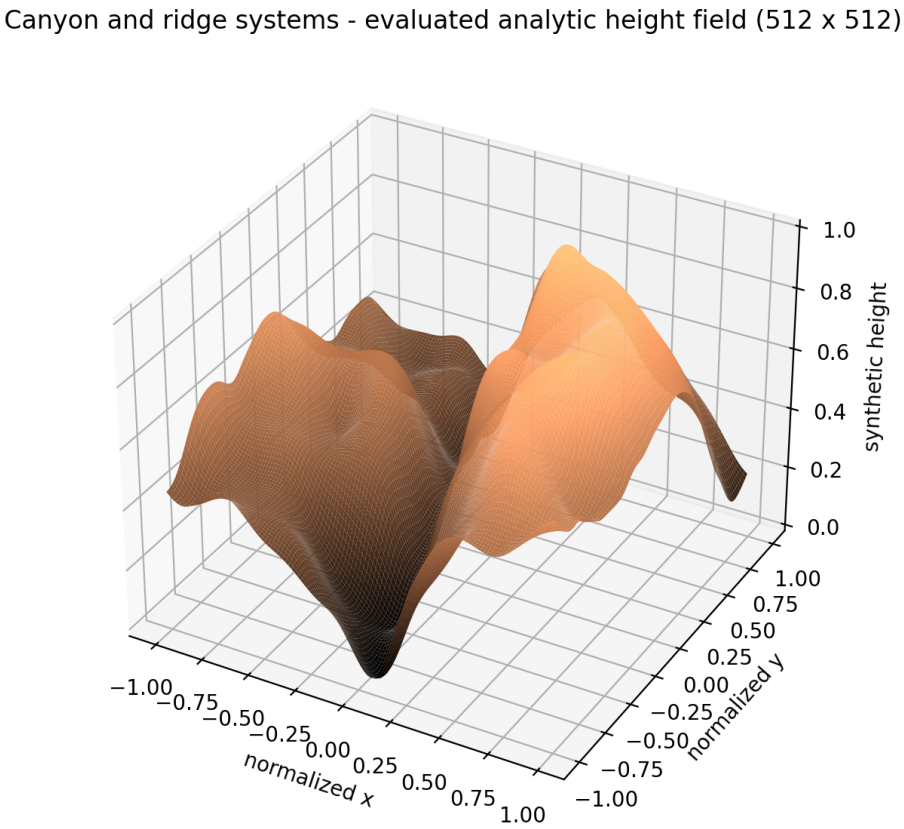


Figure 3. Actual 512 x 512 paceval height field, 4,096 features with a diagonal erosional-looking incision.

A ridge and canyon shape is encoded directly as a closed-form expression. No sediment transport or physical incision law is solved. The export preserves the mathematical output as an inspectable 3D surface.

5. Terrain contours and water-level geometry

Contour maps provide a 2D check that important topographic features are present in the actual computed data. Blue contours mark declared constant water levels; they are not predictions of physically stable water bodies.

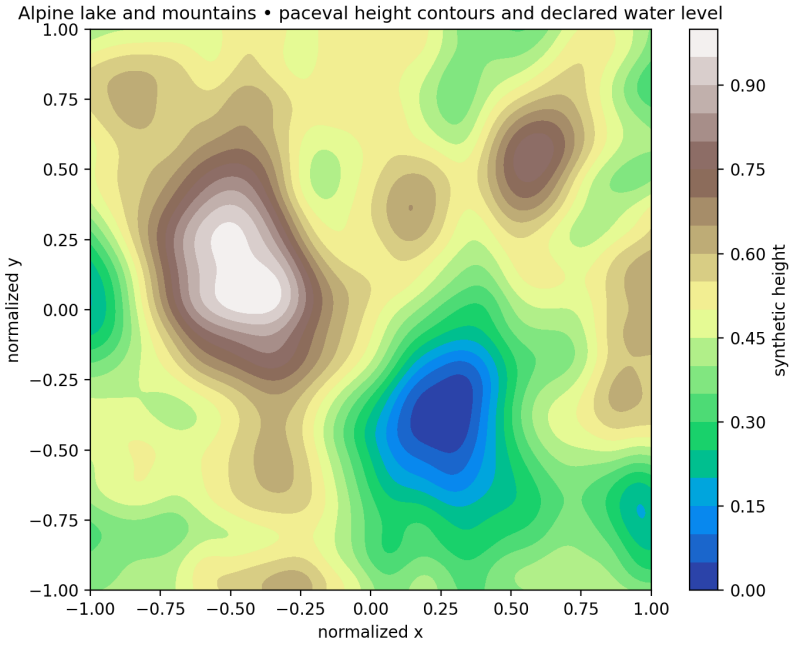


Figure 4. Alpine height contours derived directly from paceval results.

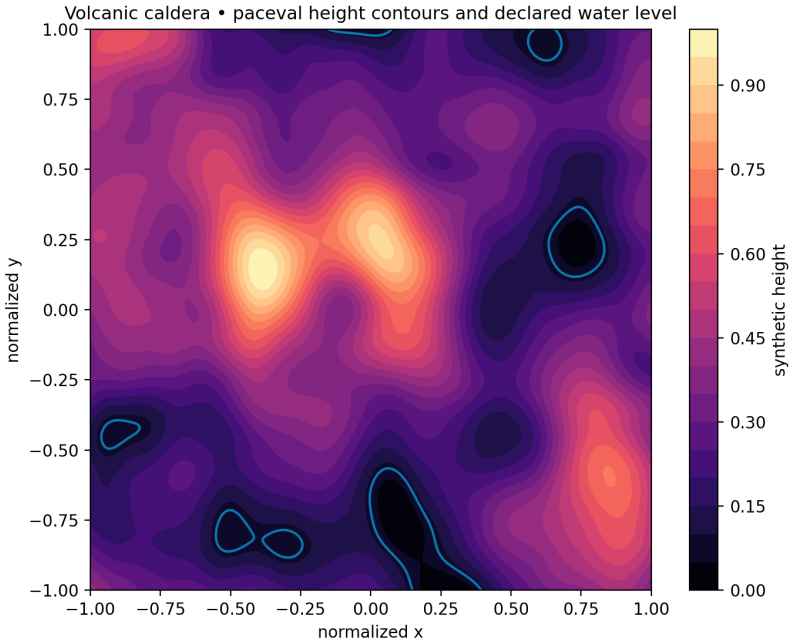


Figure 5. Caldera contours and synthetic water-level crossing.

6. Canyon contours and exported geometry

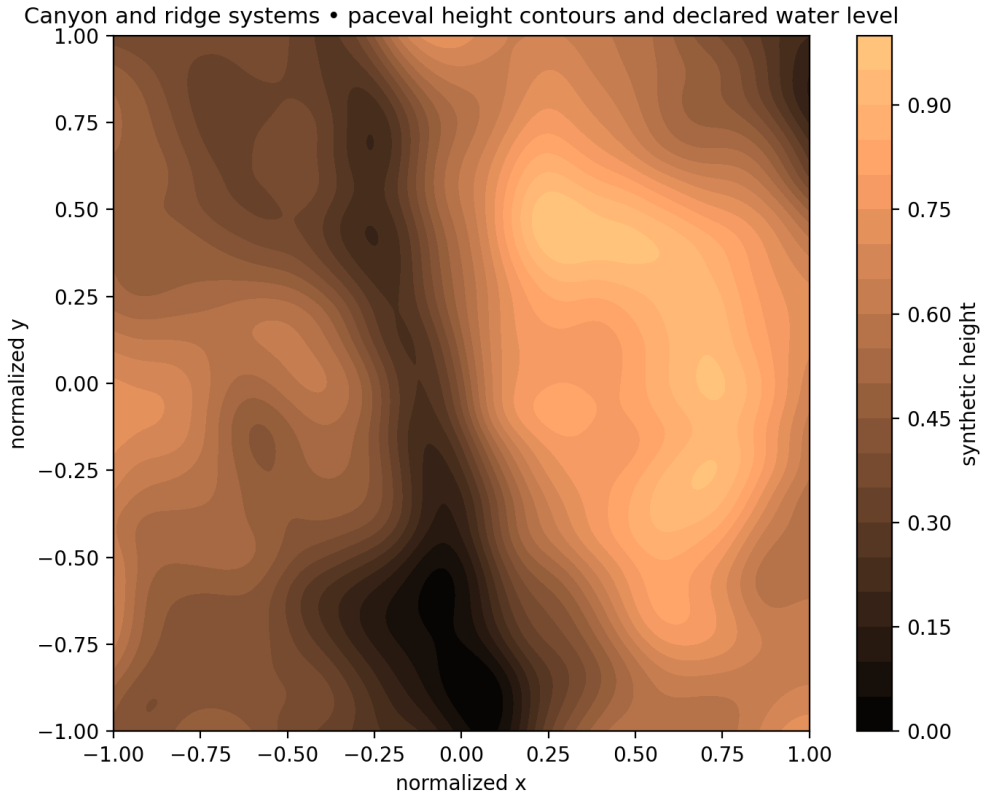


Figure 6. Contours of the actual canyon-ridge mathematical surface and a declared illustrative level.

Every 512 x 512 terrain grid produces 262,144 vertices and 522,242 triangles. The separately exported water meshes are made of constant-height triangles over cells with all four corners below the water threshold. This simple construction is not a connected-basin hydrology algorithm and must not be interpreted as a correctly clipped or physically validated shoreline.

GLB scene files were reloaded with trimesh, and the presence of nonempty triangle geometry was verified. The output is 2.5D: it has no caves, overhangs or volumetric strata.

7. Explicit expression scale and verification

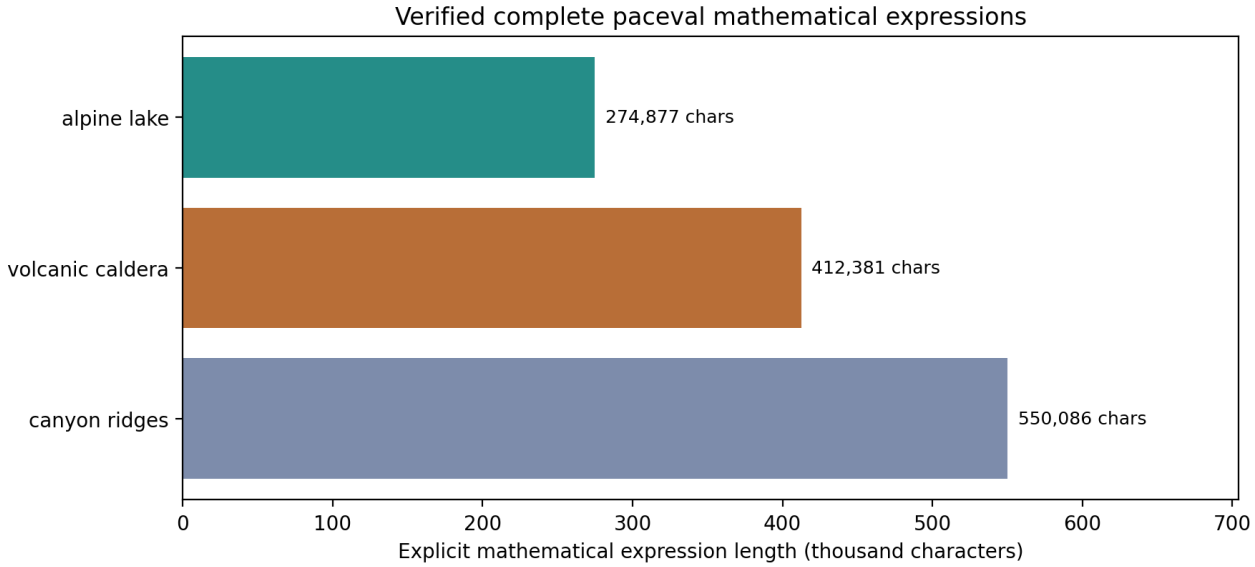


Figure 7. Verified expression lengths from the executed C++/paceval run, not projected or simulated limits.

All three full expressions were successfully created as individual computation graphs. Sampled monolithic evaluation, factorized grid reconstruction, and complete grouped evaluation at 64 checkpoints per landscape passed the predeclared numerical checks. Nine independent high-precision spot checks used a separately implemented Python coefficient generator.

What “no fixed formula-length limit” means

paceval describes no predefined product-level limit on expression length, subject to hardware and development-environment resources [4,5]. This is not a proof that arbitrary-length formulas can always be executed. In particular, one billion characters would require sufficient memory for the source string, parsing, graph representation, evaluation buffers and practical compute resources, and it has NOT been tested in this research.

The largest **actually tested complete formula** in this new experiment is 550,086 characters. The attached source whitepaper reports a separate prior test reaching 3,032,517 characters with 16,384 terms. The present report does not repeat or reclassify that earlier test as a newly executed result.

8. Research interpretation and next steps

The significant demonstrated capability is management and execution of very large editable mathematical landscape expressions, including exact formula export, repeated graph evaluation, independent numeric checking and genuine 3D geometry extraction. This does not demonstrate any CPU, GPU, energy or end-to-end rendering advantage over alternative terrain algorithms.

Meaningful industrial follow-up

To evaluate geospatial relevance, fit the analytic basis to a public digital elevation tile, preserve coordinate reference systems and vertical datums, use spatially held-out test blocks and compare elevation, slope, curvature and hydrological connectivity. For a production digital twin, integrate level-of-detail, tiling and data provenance. For physically meaningful landscape formation, couple the mathematical function to an erosion/hydrology solver rather than treating visual shapes as physics.

Reproduction

Configure CMake against a compatible separately acquired paceval Linux runtime. Run the executable with --out . --grid 512, then run the independent high-precision script and scientific figure/mesh generator. Exact build commands and a verified file inventory are documented in the root README.md. No restricted shared-library binary is redistributed.

References

[1] Physically-based analytical erosion for fast terrain generation (2024), Computer Graphics Forum. doi:10.1111/cgf.15033.

[2] USGS 3D Elevation Program standards: https://www.usgs.gov/ngp-standards-and-specifications/3d-elevation-program-standards-and-specifications

[3] OGC 3D Tiles 1.1: https://www.ogc.org/standards/3DTiles/

[4] paceval Product Brief: https://paceval.com/product-brief/

[5] paceval API and C/C++ interface: https://paceval.com/api/

Conclusion. Three new terrain expressions of more than 250,000 characters each were generated, parsed and independently evaluated with the real paceval engine; their calculated heights produced three verifiable 3D terrain scenes. Claims for larger untested formulas remain conditional on available hardware and software resources.